Lecture 11

# Scattering states: transmission, reflection, tunnelling

**Objectives.** *For the compactly supported potentials of Lecture 10, scattering solutions have amplitudes $r$ and $t$ that satisfy $|r|^2 + |t|^2 = 1$. They are not states. Packets are. We build their evolution from the free one, without the spectral theorem. For the $\delta$ Hamiltonian, we prove that it is self-adjoint and find its level and its transmission. Wave operators connect packets to the stationary amplitudes.*

- Compute reflection and transmission from the current.
- Explain tunnelling through a finite barrier.
- State the potential and momentum-support hypotheses of the packet transmission limit.

## 11.1 Waves that do not decay

Let $V$ be a potential that vanishes outside $[-R, R]$ (Definition 10.1), and let $E = \hbar^2 k^2/2m$ with $k > 0$. Outside $[-R, R]$ the equation (10.1) reads $\psi'' = -k^2\psi$, and its solutions there are the combinations of $e^{\mathrm{i}kx}$ and $e^{-\mathrm{i}kx}$ (§ 10.3), waves of momenta $\hbar k$ and $-\hbar k$. Lecture 10 solved the equation only where it could write the solutions down. In general they exist.

**Lemma 11.1** (Existence). *Let $V$ be a potential, $E$, $x_0$ real and $a, b \in \mathbb{C}$. There is exactly one continuously differentiable function $\psi$ on $\mathbb{R}$ that is twice differentiable where $V$ is continuous, satisfies (10.1) there, and has $\psi(x_0) = a$ and $\psi'(x_0) = b$.*

*Proof.* Uniqueness is Lemma 10.6. Put $g = (2m/\hbar^2)(V - E)$ and $C = \sup|g|$, let $\psi_0(x) = a + b(x - x_0)$, and

$$\psi_{n+1}(x) = a + b(x - x_0) + \int_{x_0}^{x}\int_{x_0}^{s} g(y)\,\psi_n(y)\,\mathrm{d}y\,\mathrm{d}s.$$

Fix $L > 0$, and let $M$ be the largest value of $|\psi_1 - \psi_0|$ on $I = [x_0 - L, x_0 + L]$. Since the double integral of $|y - x_0|^{2n}$ is $|x - x_0|^{2n+2}/(2n+1)(2n+2)$, induction gives $|\psi_{n+1} - \psi_n| \le MC^n\,|x - x_0|^{2n}/(2n)!$ on $I$. Since $\sum_n MC^nL^{2n}/(2n)! = M\cosh(\sqrt{C}\,L)$ is finite, the $\psi_n$ converge uniformly on $I$, and their limit $\psi$ satisfies the same equation without the index. The inner integral is continuous in $s$, so $\psi$ is continuously differentiable, with $\psi'(x) = b + \int_{x_0}^{x} g\psi\,\mathrm{d}y$, and $\psi'' = g\psi$ where $V$ is continuous. This is (10.1), and $\psi(x_0) = a$, $\psi'(x_0) = b$. Since $L$ is arbitrary, $\psi$ is defined on $\mathbb{R}$. ∎

**Proposition 11.2** (Scattering solutions). *Let $V$ be a potential that vanishes outside $[-R, R]$, and let $k > 0$. There is exactly one solution $\psi_k$ of (10.1) with $E = \hbar^2k^2/2m$, as in*

*Lemma 11.1, such that for some numbers $r$ and $t$*

$$\psi_k(x) = e^{\mathrm{i}kx} + r\,e^{-\mathrm{i}kx} \quad \text{for } x \le -R, \qquad \psi_k(x) = t\,e^{\mathrm{i}kx} \quad \text{for } x \ge R. \tag{11.1}$$

*They satisfy $|r|^2 + |t|^2 = 1$, and $t \neq 0$.*

*Proof.* Let $\varphi$ be the solution with $\varphi(R) = e^{\mathrm{i}kR}$ and $\varphi'(R) = \mathrm{i}ke^{\mathrm{i}kR}$. By uniqueness, $\varphi = e^{\mathrm{i}kx}$ for $x \ge R$, and $\varphi = ae^{\mathrm{i}kx} + be^{-\mathrm{i}kx}$ for $x \le -R$, for some $a$ and $b$. Since $V$ and $E$ are real, $\overline{\varphi}$ is a solution too, and as in Theorem 10.7 the Wronskian $W = \overline{\varphi}\varphi' - \overline{\varphi}'\varphi$ is constant. It is $2\mathrm{i}k$ for $x \ge R$ and $2\mathrm{i}k(|a|^2 - |b|^2)$ for $x \le -R$. So $|a|^2 - |b|^2 = 1$, and $a \neq 0$. Then $\psi_k = \varphi/a$ has the form (11.1) with $t = 1/a$ and $r = b/a$, and $|r|^2 + |t|^2 = (|b|^2 + 1)/|a|^2 = 1$. If $\hat{\psi}$ is a second such solution, with $\hat{r}$ and $\hat{t}$, then $d = \psi_k - \hat{\psi}$ is $(t - \hat{t})e^{\mathrm{i}kx}$ on the right and $(r - \hat{r})e^{-\mathrm{i}kx}$ on the left, and its Wronskian $\overline{d}d' - \overline{d}'d$ is $2\mathrm{i}k\,|t - \hat{t}|^2$ on the right and $-2\mathrm{i}k\,|r - \hat{r}|^2$ on the left. So both vanish, $d = 0$ for $x \ge R$, and $d = 0$ by uniqueness. ∎

The numbers $r = r(k)$ and $t = t(k)$ are the *reflection* and *transmission amplitudes*. Since $|\psi_k| = |t| > 0$ for $x \ge R$, $\psi_k$ is not square integrable, as Theorem 10.7 requires. So it is not a state. Cut off far to the right, it gives, up to the factor $t$, the Weyl sequences of the proof of Theorem 10.10, with $p_0 = \hbar k$. What it describes is read from its *current* $j = (\hbar/m)\,\mathrm{Im}(\overline{\psi}\psi')$, which is $(\hbar/2m\mathrm{i})W$ and so constant. It is $(\hbar k/m)(1 - |r|^2)$ on the left and $(\hbar k/m)\,|t|^2$ on the right. Read $e^{\mathrm{i}kx}$ as a stream of particles of velocity $\hbar k/m$. Then, of the incoming current, the fraction $|r|^2$ is reflected and $|t|^2$ is transmitted. § 11.4 says what this means for a state.

**Self-check 11.1**

1. Show that $|\psi_k(x)|^2 = 1 + |r|^2 + 2\,\mathrm{Re}(re^{-2\mathrm{i}kx})$ for $x \le -R$, and find its largest and smallest values.

## 11.2 The $\delta$ potential

Let $\alpha > 0$, and make the square well of § 10.4 narrow and deep with $\int V = -2wV_0 = -\alpha$ fixed. Then $\gamma \to 0$, so for small $w$ there is one level, and it tends to $-m\alpha^2/2\hbar^2$ (Exercise Sheet 11).

The limit is the *$\delta$ well* $-\alpha\delta$, an attraction at the single point 0. It is not a potential in the sense of Definition 10.1, and its Hamiltonian needs its own domain. We build it from the free kernel. Fix $\eta > 0$, $z = -\hbar^2\eta^2/2m$ and the kernel $G$ of Proposition 6.11. By Lemma 10.4, every $u \in D(H_0)$ is a continuous function, so $u(0)$ makes sense.

**Definition 11.3** (The $\delta$ Hamiltonian)**.** Let $\alpha$ be real. The Hamiltonian $H_\alpha$ has the domain

$$D(H_\alpha) = \{\psi = u + cG : \ u \in D(H_0),\ c \in \mathbb{C},\ c = \alpha\psi(0)\},$$

and acts by $H_\alpha\psi = H_0u + zcG$.

**Lemma 11.4** (The domain)**.** *The decomposition $\psi = u + cG$ is unique, and $D(H_\alpha)$ and $H_\alpha$ do not depend on $\eta$.*

*Proof.* By Lemma 10.4, $\langle G, (H_0 - z)v\rangle = v(0)$ for every $v \in D(H_0)$, since $G$ is real and even. If $G$ lay in $D(H_0)$, then $v(0) = \langle (H_0 - z)G, v\rangle$, and $v(x) = f(nx)$, with $f$ smooth, vanishing outside a bounded interval and $f(0) = 1$, would give $1 \le \|(H_0 - z)G\|\,\|f\|\,n^{-1/2}$ for every $n$. So $G \notin D(H_0)$, and the decomposition is unique. Let $\eta' > 0$, $z' = -\hbar^2\eta'^2/2m$, and write $G_\eta$, $G_{\eta'}$

for the two kernels. A direct integration, by Lemma 10.4, gives $(z'-z)(H_0-z')^{-1}G_\eta = G_{\eta'}-G_\eta$, so $G_\eta - G_{\eta'} \in D(H_0)$. With $u' = u + c(G_\eta - G_{\eta'})$, $\psi = u' + cG_{\eta'}$, and since $(H_0 - z')(G_{\eta'} - G_\eta) = (z' - z)G_\eta$, we get $H_0u' + z'cG_{\eta'} = H_0u + zcG_\eta$. ∎

Since $G'(0+) - G'(0-) = -2m/\hbar^2$, a function $\psi = u + cG$ of the domain with a continuously differentiable $u$ is continuous, and its slope jumps at 0 by

$$\psi'(0+) - \psi'(0-) = -\frac{2mc}{\hbar^2} = -\frac{2m\alpha}{\hbar^2}\,\psi(0). \tag{11.2}$$

Formally, integrating $-(\hbar^2/2m)\psi'' - \alpha\delta\psi = E\psi$ over a small interval around 0 gives this condition. For a smooth $u$, $H_\alpha\psi = -(\hbar^2/2m)\psi''$ away from 0, since $G'' = \eta^2 G$ there.

**Theorem 11.5** (The $\delta$ potential). *Let $\alpha$ be real. Then $H_\alpha$ is self-adjoint, and $\sigma_{\mathrm{ess}}(H_\alpha) = [0,\infty)$. If $\alpha \le 0$, then $\sigma(H_\alpha) = [0,\infty)$. If $\alpha > 0$, then $H_\alpha$ has exactly one eigenvalue,*

$$E = -\frac{m\alpha^2}{2\hbar^2},$$

*which is simple and isolated, with the eigenvector $e^{-\mu|x|}$, $\mu = m\alpha/\hbar^2$, and $\sigma(H_\alpha) = \{E\} \cup [0,\infty)$.*

*Proof.* First, $H_\alpha$ is symmetric. For $\psi_j = u_j + c_jG$ in $D(H_\alpha)$, $(H_\alpha - z)\psi_j = (H_0 - z)u_j$. With $\langle G, (H_0 - z)u_2\rangle = u_2(0)$, $u_2(0) = \psi_2(0) - c_2G(0)$ and $\overline{c_1} = \alpha\overline{\psi_1(0)}$,

$$\langle \psi_1, (H_\alpha - z)\psi_2\rangle = \langle u_1, (H_0 - z)u_2\rangle + \alpha\overline{\psi_1(0)}\psi_2(0) - G(0)\overline{c_1}c_2.$$

Since $H_0$ is symmetric and $z$, $\alpha$ and $G(0)$ are real, exchanging $\psi_1$ and $\psi_2$ gives the complex conjugate. So $\langle \psi_1, H_\alpha\psi_2\rangle = \langle H_\alpha\psi_1, \psi_2\rangle$. The domain is dense. It contains the smooth functions that vanish outside a bounded interval and near 0 (with $c = 0$). A smooth $\varphi$ that vanishes outside a bounded interval differs from such a function, $\varphi(1 - \theta(x/\epsilon))$ with $\theta$ smooth, $0 \le \theta \le 1$, equal to 1 near 0 and vanishing outside $[-2, 2]$, by at most $\sup|\varphi|\ (4\epsilon)^{1/2}$ in norm, and by S4 these $\varphi$ are dense.

Next, let $\lambda = -\hbar^2\mu^2/2m < 0$, with $\mu > 0$, and write the domain with $G_\mu$ (Lemma 11.4). Then $(H_\alpha - \lambda)\psi = (H_0 - \lambda)u$, so $\psi = u + cG_\mu$ solves $(H_\alpha - \lambda)\psi = \varphi$ exactly when $u = (H_0 - \lambda)^{-1}\varphi$ and $c = \alpha(u(0) + cG_\mu(0))$, that is,

$$\big(1 - \alpha G_\mu(0)\big)c = \alpha u(0). \tag{11.3}$$

If $\alpha G_\mu(0) \ne 1$, this has exactly one solution, and $|u(0)| \le \|G_\mu\|\,\|\varphi\|$ (Lemma 10.4), so $\psi$ is a bounded function of $\varphi$ and $\lambda$ lies in the resolvent set. Since $\alpha G_\mu(0) = m\alpha/\hbar^2\mu$, this holds for every $\lambda < 0$ if $\alpha \le 0$, and for every $\lambda < 0$ but one if $\alpha > 0$. The proof of Proposition 6.7, with one such real $\lambda$ in place of $z$ and $\overline{z}$, shows that $H_\alpha$ is self-adjoint.

If $\alpha > 0$ and $\mu = m\alpha/\hbar^2$, then (11.3) holds with $u = 0$ and every $c$. So $G_\mu \in D(H_\alpha)$ and $H_\alpha G_\mu = \lambda G_\mu$. So the number $\lambda = -m\alpha^2/2\hbar^2$ is an eigenvalue, with an eigenvector $G_\mu$, a multiple of $e^{-\mu|x|}$. Every eigenvector $u + cG_\mu$ has $(H_0 - \lambda)u = 0$, so $u = 0$, and the level is simple. The other negative numbers lie in the resolvent set, so it is isolated.

No eigenvalue is $\ge 0$. Let $H_\alpha\psi = \lambda\psi$ with $\lambda \ge 0$ and $\psi = u + cG$. Then $(H_0 - z)u = (H_\alpha - z)\psi = (\lambda - z)\psi$, a continuous function, so by the second part of Lemma 10.4 $u$ is twice differentiable, with $-(\hbar^2/2m)u'' - zu = (\lambda - z)\psi$. Since $-(\hbar^2/2m)G'' = zG$ off 0, $-(\hbar^2/2m)\psi'' = \lambda\psi$ on each half-line, and as in the first paragraph of the proof of Theorem 10.7, a square-integrable solution vanishes there. So $\psi = 0$.

Finally, the packets $\psi_n$ of the proof of Theorem 10.10, with $R = 1$, vanish near 0, so $\psi_n \in D(H_\alpha)$ with $c = 0$ and $H_\alpha\psi_n = H_0\psi_n$. As there, $[0,\infty) \subseteq \sigma_{\mathrm{ess}}(H_\alpha)$, and $\sigma(H_\alpha) \subseteq \mathbb{R}$ by Theorem 6.5. ∎

The bound state is the free kernel $G$ at its own level, and the level condition reads $\alpha G(0) = 1$. Albeverio et al. describe the same domain (Thm I.3.1.3) and state the theorem (Thms I.3.1.1 and I.3.1.4), for their $-\Delta_{\alpha',0}$ in units with $\hbar^2/2m = 1$ and $\alpha' = -2m\alpha/\hbar^2$. The square wells converge to $H_\alpha$ in the norm-resolvent sense (Thm I.3.2.3): for each nonreal spectral parameter, the resolvents converge in operator norm.

With the other sign, the *δ barrier* $\alpha\delta$, with $\alpha > 0$, is $H_{-\alpha}$, with the jump $+(2m\alpha/\hbar^2)\psi(0)$. A wave $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ for $x < 0$ and $te^{\mathrm{i}kx}$ for $x > 0$ is continuous when $1 + r = t$, and has this jump when $\mathrm{i}kt - \mathrm{i}k(1 - r) = (2m\alpha/\hbar^2)t$. So

$$t = \frac{1}{1 + \mathrm{i}m\alpha/\hbar^2 k}, \qquad r = t - 1, \qquad |t|^2 = \frac{1}{1 + m\alpha^2/2\hbar^2 E},$$

and $|r|^2 + |t|^2 = 1$. Slow particles are reflected and fast ones pass. The $\delta$ well has the same $|t|^2$.

**Self-check 11.2**

1. An electron sits in a $\delta$ well with $\alpha = 1\,\mathrm{eV\,nm}$. Find its level and the decay length $1/\eta$. Use $\hbar c = 197.3\,\mathrm{eV\,nm}$ and $mc^2 = 511\,\mathrm{keV}$.

## 11.3 The step

A potential that does not vanish on one side changes the wave number there. Let $V_0 > 0$, and let $V = V_0$ on $(0,\infty)$ and $V = 0$ on $(-\infty, 0]$. It is bounded and real, so the proof of Proposition 10.2 shows that $H_0 + V$ is self-adjoint on $D(H_0)$. The proofs of Lemmas 10.6 and 11.1 use only that $V$ is real, bounded and continuous except at finitely many jumps, so they apply too.

For $E > V_0$ let $k = \sqrt{2mE}/\hbar$ and $k' = \sqrt{2m(E - V_0)}/\hbar$, and match $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ for $x < 0$ to $te^{\mathrm{i}k'x}$ for $x > 0$ (Exercise Sheet 11). The current is $(\hbar k/m)(1 - |r|^2)$ on the left and $(\hbar k'/m)\,|t|^2$ on the right, since the transmitted particles move with the velocity $\hbar k'/m$. So the reflected and transmitted fractions of the current are

$$R = \left(\frac{k - k'}{k + k'}\right)^2, \qquad T = \frac{k'}{k}\,|t|^2 = \frac{4kk'}{(k + k')^2}, \qquad R + T = 1.$$

The transmitted fraction is not $|t|^2$, since the current weights the amplitude by the velocity.

For $0 < E < V_0$, the wave on the right is a multiple of $e^{-qx}$, with $q = \sqrt{2m(V_0 - E)}/\hbar$. Its current is 0, so the current $(\hbar k/m)(1 - |r|^2)$ on the left vanishes too, and $|r| = 1$, $R = 1$. Everything is reflected. Yet the wave does not vanish for $x > 0$. It decays over the length $1/q$, and a packet enters the step for a while before it returns. This section is stationary. § 11.4 gives the fractions their meaning for packets, for potentials that vanish outside a bounded interval.

**Self-check 11.3**

1. Show that $T \to 0$ as $E$ falls to $V_0$, and $T \to 1$ as $E \to \infty$.

## 11.4 Packets and wave operators

A state is a packet, and it moves by the evolution of $H = H_0 + V$. Theorem 7.5 builds that evolution with the spectral theorem. For a potential it can be built from the free evolution $U_t^0$ of (7.3) alone. If $U_t$ were known, $W_t = U_{-t}^0 U_t$ would satisfy $\mathrm{i}\hbar\, \mathrm{d}W_t/\mathrm{d}t = B_t W_t$, with $B_t$ below. We solve this equation instead.

**Proposition 11.6** (The evolution with a potential). *Let $V$ be a potential and $H = H_0 + V$, and put $B_s = U_{-s}^0 V U_s^0$. For every $\psi$ there is exactly one continuous solution $t \mapsto W_t\psi$ of*

$$W_t\psi = \psi - \frac{\mathrm{i}}{\hbar}\int_0^t B_s W_s \psi \,\mathrm{d}s, \tag{11.4}$$

*and $U_t = U_t^0 W_t$ is a strongly continuous unitary group with the generator $H/\hbar$. So for these $H$ the group of Theorem 7.5 exists without the spectral theorem: it is $U_t = e^{-\mathrm{i}tH/\hbar}$.*

*Proof.* Let $v = \sup|V|$. Then $\|B_s\| \le v$, and $B_s\chi$ is continuous in $s$ for every $\chi$, since $U_s^0$ is strongly continuous and keeps norms. So $B_s g(s)$ is continuous for every continuous $g$, since $\|B_{s'}g(s') - B_s g(s)\| \le v\,\|g(s') - g(s)\| + \|(B_{s'} - B_s)g(s)\|$.

*Existence.* Put $W_t^0\psi = \psi$, and let $W_t^{n+1}\psi$ be the right side of (11.4) with $W_s^n\psi$ in the integral. Each is continuous in $t$ (Lemma 7.7), and by the norm bound of Lemma 7.7 and induction, $\left\|W_t^{n+1}\psi - W_t^n\psi\right\| \le \|\psi\|\,(v\,|t|\,/\hbar)^{n+1}/(n+1)!$. So the $W_t^n\psi$ converge uniformly for $|t| \le \tau$, for every $\tau$, and the limit $W_t\psi$ is continuous, linear in $\psi$ and bounded by $e^{v|t|/\hbar}\,\|\psi\|$. By the norm bound again, the integrals converge too, so $W_t\psi$ solves (11.4).

*Uniqueness.* Let $a$ be real, and let $\chi$ be continuous with $\chi(t) = -(\mathrm{i}/\hbar)\int_0^t B_{s+a}\chi(s)\,\mathrm{d}s$. If $M$ is the largest value of $\|\chi\|$ on $[-\tau,\tau]$, induction gives $\|\chi(t)\| \le M(v\,|t|\,/\hbar)^n/n!$ there, so $\chi = 0$. With $a = 0$, this is the uniqueness claimed.

*Norms.* Let $f(t) = W_t\psi$. By the last claim of Lemma 7.7, $f$ is differentiable with $f'(t) = -(\mathrm{i}/\hbar)B_t f(t)$. So, by the product rule for $\langle f(t), f(t)\rangle$, the derivative of $\|f(t)\|^2$ is

$$2\,\mathrm{Re}\,\langle f(t), f'(t)\rangle = (2/\hbar)\,\mathrm{Im}\,\langle f(t), B_t f(t)\rangle = 0,$$

since $B_t$ is symmetric. So $\|f(t)\|^2$ is constant, and $\|W_t\psi\| = \|W_0\psi\| = \|\psi\|$.

*The group.* Fix $s$, and put $\varphi = U_s^0 W_s\psi$. Since $U_{-s}^0 B_r = B_{r+s}U_{-s}^0$, splitting the integral for $W_{t+s}\psi$ at $s$ shows that $\chi(t) = W_{t+s}\psi - U_{-s}^0 W_t\varphi$ satisfies the equation of the uniqueness step with $a = s$. So $\chi = 0$, that is, $U_{t+s}\psi = U_t U_s\psi$. Each $U_t$ keeps norms and $U_t U_{-t} = U_0 = \mathbb{1}$, so $U_t$ is unitary, and $U_t\psi$ is continuous in $t$, by the bound used for $B_s g(s)$.

*The generator.* For $\psi \in D(H_0)$,

$$\frac{U_h\psi - \psi}{h} = U_h^0\,\frac{W_h\psi - \psi}{h} + \frac{U_h^0\psi - \psi}{h} \longrightarrow -\frac{\mathrm{i}}{\hbar}(V + H_0)\psi,$$

since $(W_h\psi - \psi)/h \to -(\mathrm{i}/\hbar)B_0\psi = -(\mathrm{i}/\hbar)V\psi$ by (11.4), and the $U_h^0$ keep norms and tend to $\mathbb{1}$ on every vector. So the generator extends $H/\hbar$, which is self-adjoint (Proposition 10.2), and as in the proof of Theorem 7.4 it equals $H/\hbar$. By Proposition 7.9, every group with the generator $H/\hbar$ is $(U_t)$. ∎

The iterates $W_t^n$ are the partial sums of the *Dyson series*. A packet that comes in from far away should move, at early times, like a free packet $U_t^0\psi$. The wave operators compare the two evolutions.

**Theorem 11.7** (Wave operators). *Let $V$ be a potential. For every $\psi \in L^2(\mathbb{R})$ the limits*

$$\Omega_{\text{in}}\psi = \lim_{t\to-\infty} U_t^\dagger U_t^0 \psi, \qquad \Omega_{\text{out}}\psi = \lim_{t\to+\infty} U_t^\dagger U_t^0 \psi$$

*exist in norm, and $\|\Omega_{\text{in}}\psi\| = \|\Omega_{\text{out}}\psi\| = \|\psi\|$. Every eigenvector of $H$ is orthogonal to $\Omega_{\text{in}}\psi$ and to $\Omega_{\text{out}}\psi$.*

*Proof.* We treat $\Omega_{\text{out}}$; $\Omega_{\text{in}}$ is the same with $t \to -\infty$. Write $f_\psi(t) = U_t^\dagger U_t^0 \psi = U_{-t} U_t^0 \psi$. Each $U_{-t}U_t^0$ is unitary, so $\|f_\psi(t) - f_\chi(t)\| = \|\psi - \chi\|$, and it suffices to prove convergence for $\psi$ in a dense set. Let $0 < a < b$ and let $\tilde\psi$ be smooth and vanish outside $[a,b]$, or outside $[-b,-a]$. By S1 and S4, and the cut near 0 of the proof of Theorem 11.5, the sums of such $\psi$ are dense.

*Decay.* By (7.3) and the inversion formula,

$$(U_t^0\psi)(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{\mathrm{i}\Phi(p)/\hbar} \tilde\psi(p)\,\mathrm{d}p, \qquad \Phi(p) = px - \frac{p^2 t}{2m}.$$

Let $V$ vanish outside $[-R,R]$. For $|x| \le R$, $a \le p \le b$ and $|t| \ge 2mR/a$, $|\Phi'(p)| = |x - pt/m| \ge a\,|t|\,/2m$. Write $e^{\mathrm{i}\Phi/\hbar} = (\hbar/\mathrm{i}\Phi')(e^{\mathrm{i}\Phi/\hbar})'$ and integrate by parts twice, with no boundary terms. The integral becomes $-\hbar^2$ times the integral of $e^{\mathrm{i}\Phi/\hbar}$ multiplied by

$$\left(\frac{(\tilde\psi/\Phi')'}{\Phi'}\right)' = \frac{(\tilde\psi)''}{\Phi'^2} + \frac{3t(\tilde\psi)'}{m\Phi'^3} + \frac{3t^2\tilde\psi}{m^2\Phi'^4}.$$

Here $\Phi'' = -t/m$. The derivatives of $\tilde\psi$ are bounded on their fixed support. Since $|\Phi'| \ge a\,|t|\,/2m$, each term is bounded by a constant times $1/t^2$. So $\left|(U_t^0\psi)(x)\right| \le C/t^2$ on $[-R,R]$ for large $|t|$, and $\left\|VU_t^0\psi\right\| \le \sup|V|\ (2R)^{1/2} C/t^2$.

*Cook's argument.* The vector $\psi$ lies in $D(H_0) = D(H)$, and so does $U_t^0\psi$ (Proposition 7.3). Divide

$$f_\psi(t+h) - f_\psi(t) = U_{-t-h}\big(U_{t+h}^0\psi - U_t^0\psi\big) + \big(U_{-t-h} - U_{-t}\big)U_t^0\psi$$

by $h$. Since $\|U_{-t-h}\| = 1$ and $U_{-t-h}$ tends to $U_{-t}$ on every vector, the first term tends to $-(\mathrm{i}/\hbar)U_{-t}H_0U_t^0\psi$, and by Propositions 7.3 and 11.6 the second to $(\mathrm{i}/\hbar)U_{-t}HU_t^0\psi$. So $f_\psi'(t) = (\mathrm{i}/\hbar)U_{-t}VU_t^0\psi$, a continuous function of $t$. For every $\varphi$, the derivative of $\langle\varphi, f_\psi(t)\rangle$ is $\langle\varphi, f_\psi'(t)\rangle$. Integrating it, with $\varphi = f_\psi(t') - f_\psi(t)$, and dividing by $\|\varphi\|$,

$$\left\|f_\psi(t') - f_\psi(t)\right\| \le \frac{1}{\hbar}\int_t^{t'} \left\|VU_s^0\psi\right\|\,\mathrm{d}s \le \frac{\sup|V|\ (2R)^{1/2}C}{\hbar}\left(\frac{1}{t} - \frac{1}{t'}\right)$$

for large $t < t'$. So $f_\psi(t)$ converges as $t \to \infty$. Its norm is $\|\psi\|$ for every $t$, and so is the norm of the limit.

*Bound states.* Let $H\varphi = E\varphi$. By Propositions 7.3 and 11.6, $e^{\mathrm{i}Et/\hbar}U_t\varphi$ has the derivative 0, so it is constant and $U_t\varphi = e^{-\mathrm{i}Et/\hbar}\varphi$. Hence $\left|\langle\varphi, f_\psi(t)\rangle\right| = \left|\langle\varphi, U_t^0\psi\rangle\right|$. For $L \ge R$ the decay step on $[-L,L]$ bounds this by $\left\|1_{|x|>L}\varphi\right\|\,\|\psi\| + (2L)^{1/2}\,\|\varphi\|\,C_L/t^2$, which is small for large $L$ and then large $t$. So $\langle\varphi, \Omega_{\text{out}}\psi\rangle = 0$ for these $\psi$, and by density for all. ∎

This is Cook's method (Teschl, Lemma 12.3).

Since $U_t$ is unitary,

$$\left\|U_t\Omega_{\text{out}}\psi - U_t^0\psi\right\| = \left\|\Omega_{\text{out}}\psi - U_t^\dagger U_t^0\psi\right\| \longrightarrow 0.$$

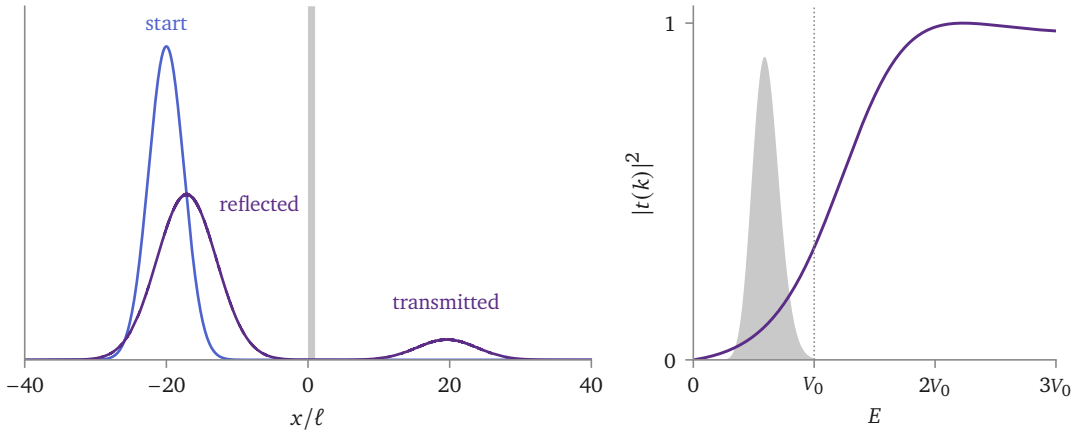



Figure 11.1: A packet at the barrier of § 11.5, of height $V_0$ and width $\ell$, with $mV_0\ell^2/\hbar^2 = 4$. Left: the position density at the start and at a later time, computed by evolving the packet; the barrier is grey. Right: $|t(k)|^2$ against the energy $E = \hbar^2k^2/2m$, and the packet's distribution of energies (grey, not to scale). The transmitted probability, 0.11, is the mean of $|t(k)|^2$ under it.

So the state $\Omega_{\text{out}}\psi$ moves at late times like the free packet $U_t^0\psi$, and $\Omega_{\text{in}}\psi$ at early times. These are states that leave, and the theorem shows that they are orthogonal to the bound states, which stay. More is true, and cited. By Enss's theorem (Teschl, Thm 12.12), every state orthogonal to all bound states is some $\Omega_{\text{in}}\psi$ and some $\Omega_{\text{out}}\chi$: *asymptotic completeness*. So the *scattering operator* $S = \Omega_{\text{out}}^\dagger\Omega_{\text{in}}$ is unitary and maps the incoming free packet $\psi$ to the outgoing one, $\psi_{\text{out}} = S\psi$. It commutes with $H_0$ (Teschl, (12.10)), so the kinetic energy is kept, and in momentum it acts by the amplitudes $r$ and $t$ (Takhtajan, Thm 2.5). With the late-time law of free packets, cited in Lecture 12, this gives the following, where the amplitude carries its argument, $t(p/\hbar)$, to keep it apart from the time $t$. For a unit vector $\varphi$ with $\tilde\varphi = 0$ on $(-\infty, 0)$, a packet incoming from the left, the probability that $U_t\Omega_{\text{in}}\varphi$ is read at $x > 0$ tends, as $t \to \infty$, to

$$\int_0^\infty |t(p/\hbar)|^2 \, |\tilde\varphi(p)|^2 \, \mathrm{d}p$$

Teta, (8.79)–(8.81), finds the same integral for a Gaussian packet far to the left of a $\delta$ barrier, up to an exponentially small error. So $|t(k)|^2$ is approximately the transmission probability of a packet whose momenta lie near $\hbar k$. Figure 11.1 shows a packet of mean energy about $0.6V_0$ at a barrier of height $V_0$: about 11% of it passes, the mean of $|t(k)|^2$ under its distribution of energies $E = \hbar^2k^2/2m$.

**Self-check 11.4**

1. Show that $\Omega_{\text{out}}U_s^0 = U_s\Omega_{\text{out}}$ for every real $s$.

## 11.5 Tunnelling

A barrier of height $V_0 > 0$ and full width $\ell$ is the potential $V = V_0$ on $(0, \ell)$ and $V = 0$ outside. A classical particle with energy $E < V_0$ is reflected. Matching at 0 and at $\ell$, as for the step (Exercise Sheet 11), gives the transmission probability

$$|t|^2 = \left[1 + \frac{V_0^2 \sinh^2(q\ell)}{4E(V_0 - E)}\right]^{-1}, \qquad q = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

For $q\ell \gg 1$, $\sinh(q\ell) \approx e^{q\ell}/2$, and

$$|t|^2 \approx \frac{16E(V_0 - E)}{V_0^2}\, e^{-2q\ell}.$$

The particle passes with a probability that falls exponentially with the width, at the rate $2q$ set by the missing energy $V_0 - E$. This is *tunnelling*. A transmitted particle does not lose energy in the barrier. Its wave on the right is $te^{\mathrm{i}kx}$, with the incoming $k$, and $S$ commutes with $H_0$. Only the probability is reduced. Gamow explained $\alpha$ decay in this way in 1928. The $\alpha$ particle tunnels out through the Coulomb barrier of the nucleus, and a factor like $e^{-2q\ell}$, for a barrier of varying height, spreads the lifetimes over more than twenty orders of magnitude (Exercise Sheet 11).

**Self-check 11.5**

1. For $V_0 - E = 1\,\mathrm{eV}$ and an electron, by what factor does $|t|^2$ fall when the barrier widens by $0.1\,\mathrm{nm}$, for $q\ell \gg 1$?

**The physics**

In 1982 Binnig, Rohrer, Gerber and Weibel reported measurements with a tungsten tip held a fraction of a nanometre from a platinum plate, with a gap they could set: the tunnelling current through the vacuum falls exponentially with the width of the gap. The same year they scanned such a tip over a surface at a fixed current, tracing its height: the scanning tunnelling microscope. In the model of this section, an electron that lacks $V_0 - E = 4\,\mathrm{eV}$, of the order of the energy needed to free an electron from a metal, has $q \approx 10\,\mathrm{nm}^{-1}$, and a gap wider by $0.1\,\mathrm{nm}$ cuts the transmission by $e^{-2q\times 0.1\,\mathrm{nm}} \approx 1/8$.

**Scattering**

*States:* the scattering solutions $\psi_k$ (Proposition 11.2) are not square integrable, so they are not states; packets are. For a potential $V$ (Definition 10.1), the state $\Omega_{\mathrm{out}}\psi$ moves at late times like the free packet $U_t^0\psi$, and it is orthogonal to the bound states (Theorem 11.7). *Observables:* the $\delta$ Hamiltonian $H_\alpha$ is self-adjoint on $D(H_\alpha)$, with $\sigma_{\mathrm{ess}}(H_\alpha) = [0,\infty)$ and one level $-m\alpha^2/2\hbar^2$ for $\alpha > 0$ (Theorem 11.5). *Measurement:* by the cited theorems of § 11.4, for a potential and a unit incoming free packet $\varphi$ with $\tilde\varphi = 0$ on $(-\infty, 0)$, the state $U_t\Omega_{\mathrm{in}}\varphi$ is read at $x > 0$, as $t \to \infty$, with probability tending to $\int_0^\infty |t(p/\hbar)|^2\, |\tilde\varphi(p)|^2\, \mathrm{d}p$.

**Summary.**

- *Scattering solutions*: for a potential that vanishes outside $[-R, R]$ and $k > 0$, the equation has exactly one solution $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ on the left and $te^{\mathrm{i}kx}$ on the right, and $|r|^2 + |t|^2 = 1$ (Lemma 11.1 and Proposition 11.2).
- *The $\delta$ potential*: the Hamiltonian $H_\alpha$ of $-\alpha\delta$ is self-adjoint on $D(H_\alpha)$, with one level $-m\alpha^2/2\hbar^2$ if $\alpha > 0$ (Theorem 11.5); the barrier $\alpha\delta$ transmits $|t|^2 = 1/(1 + m\alpha^2/2\hbar^2 E)$.
- *The step*: $R = ((k - k')/(k + k'))^2$ and $T = 4kk'/(k + k')^2$ above it, $R = 1$ below it.
- *Evolution and wave operators*: for a potential, the Dyson series builds $e^{-\mathrm{i}tH/\hbar}$ (Proposition 11.6), and $\Omega_{\mathrm{in}}$ and $\Omega_{\mathrm{out}}$ exist, keep norms and miss the bound states (Theorem 11.7); asymptotic completeness and $S$ are cited.
- *Tunnelling*: a barrier transmits about $16E(V_0 - E)V_0^{-2}e^{-2q\ell}$ for $q\ell \gg 1$, with no loss of energy.

Lecture 12 asks what the states that stay and the states that leave do in time.

**Self-check answers.**

**11.1** 1. Expanding $(e^{\mathrm{i}kx} + re^{-\mathrm{i}kx})(e^{-\mathrm{i}kx} + \bar{r}e^{\mathrm{i}kx})$ gives $1 + |r|^2 + re^{-2\mathrm{i}kx} + \overline{re^{-2\mathrm{i}kx}}$. As $x$ varies, the last two terms take every value $2|r|\cos\theta$, so $|\psi_k|^2$ runs between $(1 - |r|)^2$ and $(1 + |r|)^2$.

**11.2** 1. $E = -mc^2\alpha^2/2(\hbar c)^2 = -511\,000/(2 \times 197.3^2)\,\mathrm{eV} \approx -6.56\,\mathrm{eV}$, and $1/\eta = (\hbar c)^2/(mc^2\alpha) = 197.3^2/511\,000\,\mathrm{nm} \approx 0.076\,\mathrm{nm}$.

**11.3** 1. With $s = k'/k = \sqrt{1 - V_0/E}$, $T = 4s/(1 + s)^2$. As $E$ falls to $V_0$, $s \to 0$ and $T \to 0$; as $E \to \infty$, $s \to 1$ and $T \to 1$.

**11.4** 1. Since $U_t^\dagger = U_{-t} = U_s U_{-t-s}$, $U_t^\dagger U_t^0 U_s^0 \psi = U_s U_{t+s}^\dagger U_{t+s}^0 \psi$. As $t \to \infty$, the left side tends to $\Omega_{\mathrm{out}} U_s^0 \psi$, and the right side to $U_s \Omega_{\mathrm{out}} \psi$, since $U_s$ is bounded.

**11.5** 1. $q = \sqrt{2mc^2 \times 1\,\mathrm{eV}}/\hbar c \approx 5.12\,\mathrm{nm}^{-1}$, so the factor is $e^{-2q \times 0.1\,\mathrm{nm}} \approx 0.36$, about $1/2.8$.

## References

Albeverio, Gesztesy, Høegh-Krohn and Holden, *Solvable Models in Quantum Mechanics*, Thm I.3.1.1 (p. 76), Thms I.3.1.3, I.3.1.4 (p. 78), Thm I.3.2.3 (p. 81). Teschl, *Mathematical Methods in Quantum Mechanics*, (12.10) (p. 248), Lemma 12.3 (p. 249), Thm 12.12 (p. 256). Takhtajan, *Quantum Mechanics for Mathematicians*, Thm 2.5 (p. 175). Teta, *A Mathematical Primer on Quantum Mechanics*, (8.79)–(8.81) (p. 190). Binnig, Rohrer, Gerber and Weibel, *Appl. Phys. Lett.* 40, 178 (1982), and *Phys. Rev. Lett.* 49, 57 (1982). Gamow, *Z. Phys.* 51, 204 (1928).